Engineering Journal - Yasteel Ramphal 2108498

Monday, 04 August 2025 19:57

21 July

- PRELIMINARY MEETING WITH SUPERVISOR Prof Turgay

Meeting Minutes:
Meet group members
Define Client Needs: PROFITABLE & SUSTAINABLE (
Scope: TRASH ID & SORTING (+ Aux in Required based on design needs) ONLY
Not Limited to ML Sorting methods only (Also traditional Methods i.e. Magnets for Ferous obj)

22 July

Topic 5

Topic 5

Project Title: Smart and Sustainable Waste Sorting System for Beverage Containers
Supervisor: Prof Turgay Celik
Area: Software

Students from: Only IE

Machine vision
AI SORTING
SCADA system?
Data protection
System intergration
& sharing

Pre- Requisite Courses: Machine Learning, Secure Computing, Software Development III

Project Brief:

South Africa faces a pressing waste management crisis, with beverage containers like plastic (1) bottles, glass (2), and (3) cans often ending up in landfills due to inefficient recycling and manual sorting processes. This project challenges students to design a smart and sustainable waste sorting system that can automatically recognize and separate different beverage containers, improving recycling efficiency while promoting a circular economy. FINANCIALY VIABLE!

A key innovation is enabling multiple systems to learn from each other by sharing insights to enhance sorting accuracy—without compromising data privacy. Beyond automation, students must design the entire waste journey, including intake, identification, sorting, user interaction, and integration with sustainability principles. The ultimate goal is to create an intelligent, eco-friendly system that reduces landfill waste, conserves energy, and encourages reuse, paving the way for a cleaner and more sustainable future.

Back to the list of projects

Goals/Objectives (Systems Of Interest)

S1 > Object handeling (Conveyors / Robotic & pneumatic ARMS)
S2 > Object Identification [Computer Vision - CNN & Different Wave Length i.e. XRAY (eg TSA Scanner) & Near wave IR]
S3 > Obj SORTING (Robotic ARM / Pneumatic Blower System

ST > System Intergration (S1+S2+S3 = ST)

23 July

Scope:

> Beverage containers (Plastic/Aluminium/Glass)
> Not Collection
> Not POST Sorting Packaging
> Not Cleaning
> Fixed Contract on Price of Sorted Items

Assumptions

> Deformation of objects (Squashed & Crumpled)
> Partially filled (Variance in mass/density) still contain liquid
> Recycled obj will be Re-Manufactured (i.e. not cleaned & reused)
> Some obj will contain Multiple materials (Plastic labels)
> Quality of Material will not be assessed
↳ i.e. how many Recycling Cycle each obj has left / undergone

24 July

Projet Time line

Week 1 - Design objectives definition; System Context
- Background Research (commercial Research, Algorithms!)
Week 2 - Existing Soln. ; Sensor Research, AI Systems, Sorting mech
Week 3 - AI Systems, Sensor ; SYSTEM Integration
Week 4 - Integration ; Finalise Soln
Week 5 - Presentation, Report writing
Week 6 - Report writing + SUBMISSION:
1 September @ 8:00 AM
Deliverables (1) Cover Page
(2) Technical Report (15 Pg)
(3) Nontechnical Report (2 Pg)
(4) Journal (This)

Researched: SA Waste Management
Notes: 10% Recycled (Government website)
R15 Bn Economy!
Schenck, R. & Blaauw, P. (2011) - "The Work and Lives of Street Waste Pickers in Pretoria
> Manual Sorting 60-75% Acc
> Health Hazards
(Alu R9 - Plastic R7, Glass R1)

7 HRS - Context set

High level System

I/P Sensor
↳ RGB Camera (8MP - 60FPS)
↳ NIR Spectrometer (700 - 2500 nm)
↳ Conveyor Position Sensor
↳ Enviroment Sensor

AI & DSP
↳ YOLO Obj detection
↳ CNN feature extraction
↳ Ensemble Decision Fusion

Control System
↳ Pneumatic Ejector
Conveyor Control
Sorting Bin
Reject handeling

25 July

Investigation: Computer vision approaches

> Yolo v11 Good for RT App
> Single stage fast inference
↳ 2 stage slow but Accurate
> TRANSFER LEARNING!
↳ Adaptable to new waste streams

Alternatives: · Yolo v8
· CNN
· Fast R CNN
· EfficientDet

6 hrs - Computer Vision Analysis (Yolo Comparison Analysis)


Key Features

- **Enhanced Feature Extraction:** YOLO11 employs an improved backbone and neck architecture, which enhances feature extraction capabilities for more precise object detection and complex task performance.
- **Optimized for Efficiency and Speed:** YOLO11 introduces refined architectural designs and optimized training pipelines, delivering faster processing speeds and maintaining an optimal balance between accuracy and performance.
- **Greater Accuracy with Fewer Parameters:**
- **Adaptability Across Environments:**
- **Broad Range of Supported Tasks:**

1. Seddini, M.O. & Triqui-Sari, L. (2023) - "Computer Vision Techniques for Intelligent Detection and Classification of Waste Sorting"
2. Abo-Zahhad, M.M. (2025) - "Real time intelligent garbage monitoring using YOLOv8 and YOLOv5"

26-27 July (weekend)

· EfficientDet

- **Broad Range of Supported Tasks:** Whether it's object detection, instance segmentation, image classification, pose estimation, or oriented object detection (OBB), YOLO11 is designed to cater to a diverse set of computer vision challenges.

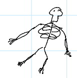
1. Seddini, M.O. & Triqui-Sari, L. (2023) - "Computer Vision Techniques for Intelligent Detection and Classification of Waste Sorting"
2. Abo-Zahhad, M.M. (2025) - "Real time intelligent garbage monitoring using YOLOv8 and YOLOv5"

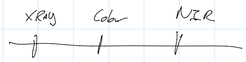
8 hrs — Computer Vision Analysis (Yolo Comparison Analysis)

26-27 July (weekend)

Sensor Analysis :

→ XRAY - Costly
- Dangerous
- Good material Comp.





→ Weight sensors - Cheap
- Little info
- half filled bottels Accuracy

✓ → NIR - Costly (less than XRAY)
- Good material Comp.
- Line of sight
- Slow

- NIR wavelength range 900-1700nm optimal for plastic classification
- PET: 98.7% accuracy at 1215nm, 1240nm wavelengths
- HDPE: 97.9% accuracy at 1168nm, 1193nm wavelengths
- Glass: 99.2% accuracy with broad 1100-1300nm absorption
- Aluminum: 99.8% accuracy with metallic reflection pattern

1. Kumar, L.M. et al. (2014) - "Design of embedded control system for efficient sorting of waste plastics using NIR Spectroscopy"
2. Rani, M. et al. (2019) - "Miniaturized Near-Infrared Spectrometer in Plastic Waste Sorting"
3. Vidal, C. & Pasquini, C. (2021) - "Comprehensive microplastics identification based on NIR hyperspectral imaging"

→ LASER - Induced
→ Mid wave NIR

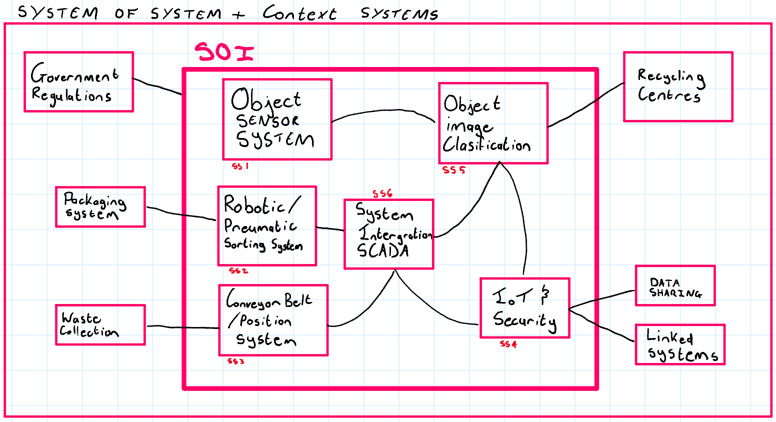
10 Hrs Sensor Comparison

28 July

Defined SOI

Monday, 11 August 2025
16:28

SYSTEM OF SYSTEM + Context SYSTEMS

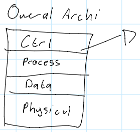


Researched - Regulations : → POPI
SANS 10366
ISA 62443

- Automation Architectures - SCADA
- Protocals for Communication
  - EtherCAT - High speed RT
  - ModBus - Conveyor
  - ProfiNET - Established Cost / Cmplx Sys

1. Babayigit, B. & Abubaker, M. (2023) - "Industrial IoT: Improvements Over Traditional SCADA Systems"
2. EtherCAT Technology Group documentation on real-time industrial Ethernet

Overal Archi



→ SCADA
↳ Master Clock - Sync
↳ HMI - Interface

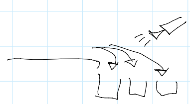
Material	Approx. Price (ZAR/kg)
Aluminum	R41.00 Scrap Metal Prices South Africa
Copper	R138.00 Scrap Metal Prices South Africa
Steel	R14.50 Scrap Metal Prices South Africa
Brass	R72.00 Scrap Metal Prices South Africa
Glass bottles	~R1/kg 2025/2026
Cardboard/paper	R0.56–R2.10/kg 2025/2026 +1
Aluminum cans (collectors)	R5–R8/kg Reddit

11 Hrs - System definition
- Architect. Design
- SCADA design

29 July

PHysical Sorting

Pneumatic - Cheap
- Complex
- Fast
- Clunky



Vaccume Gantry - Expensive
- Very High throug put (Multi-object
- Precision

Robotic ARM - Expensive
- Slow
- Precision
- Scalable (MultiStage)

Approach	Throughput	Accuracy	Flexibility	Cost
Robotic Arms	60 picks/min	High	Excellent	High
Pneumatic Ejection	200+ items/min	Moderate	Limited	Low
Magnetic Separation	Continuous	High (metals only)	Material-specific	Moderate

Robotic Arm Interface : Google RT2
: ROS 2
: Move IT
: Gripper

1. Cheng, T. et al. (2024) - "Optimizing Waste Sorting for Sustainability: AI-Powered Robotic Solution"

2. Diya, S.Z. et al. (2018) - "Developing Intelligent Waste Sorting System with Robotic Arm

7 Hrs - Mech System

30 July

1. Multi-modal sensor fusion: RGB + NIR for comprehensive material identification
2. YOLOv11 + 1D CNN architecture for real-time processing
3. Three-arm robotic configuration for throughput requirements
4. EtherCAT communication for deterministic timing contro

4 HRs colidatde design

31 July - 3 Aug

Investigated X-ray sorting, magnetic density separation, and pneumatic systems as alternative approaches.

Research Papers Reviewed:

1. Sterkens et al. (2022) - "In-line sorting system with battery detection using X-ray transmission and deep learning"
2. Wang, L. et al. (2024) - "Innovative Magnetic Density Separation Process for Sorting Granular Solid Wastes"
3. Tajfirooz, S. (2021) - "Fluid-Magnetic Solution for Sorting Plastic Waste"

Red wave System

X-ray Sorting Analysis:

Dual-energy X-ray transmission (DEXRT) enables density-based material separation
Effective for dark/flame-retardant polymers invisible to NIR
Higher equipment costs (~R2-4M additional) and radiation safety requirements
Companies: TOMRA AUTOSORT, SGM, X-SORT systems

Magnetic Density Separation (MDS):

Uses magnetized ferrofluids to create apparent density gradients
Continuous separation capability for multiple plastic types simultaneously
More efficient than traditional sink-float methods
Lower energy consumption than optical systems

Pneumatic Systems:

Fastest response times for lightweight materials
Limited to relatively light objects (<100g typically)
Songdo, South Korea implementation: 95% CO2 reduction in waste collection
Lower maintenance complexity than mechanical systems

Time : 8 hours
Tasks : Alternative technology assessment, cost-benefit analysis

4 Aug

Investigated advanced AI techniques and federated learning for waste sorting.

Research Papers Reviewed:

1. Lin, K. et al. (2023) - "MSWNet: Visual deep machine learning method using transfer learning based on ResNet 50"

2. Nahiduzzaman et al. (2025) - "Automated waste classification using deep learning techniques"
3. Casao, S. et al. (2024) - "SpectralWaste Dataset: Multimodal Data for Waste Sorting Automation"

Federated Learning Implementation:
Hub-and-spoke architecture for cross-facility learning
TrashNet dataset (2527 images) for RGB baseline
NRGlassNet dataset (1378 images) for NIR+RGB correlation
Privacy-preserving model parameter sharing only

Model Type	Accuracy	Speed	Resource Usage	Deployment Complexity
1D CNN (NIR)	99.2%	5-8ms	50K-5M parameters	Low
PLS Regression	95-99%	2-5ms	Minimal	Very Low
SVM	97.5%	1-3ms	Variable	Low
Transformer Models	99%+	50-100ms	100M+ parameters	High

Active Learning Strategy:
5% random sampling for ground truth verification
Weekly cross-validation with reference materials
Real-time confidence tracking for model retraining triggers

Time : 7 hours
Tasks : AI model selection, learning architecture design

2. Operating Expenditure (OPEX)

Breakdown for a **South African industrial setup:**

Cost Item	Estimate per Month	Notes
Electricity	~R2,500	Line-scan NIR + RGB + lights + GPU server ≈ 1.5–2.0 kW, running 4–6 hrs/day, 20 days/month, at ~R2.50/kWh
Labor	R15,000–R30,000	One skilled operator/technician for supervision and maintenance (reduced crew compared to manual sorting)
Maintenance	R5,000–R8,000	Includes calibration, lens cleaning, replacing LED arrays every years
Consumables	R500–R1,000	Cleaning supplies, compressed air for ejectors
Repairs/Service Contracts	R2,000	Average monthly allowance for unexpected downtime

Total OPEX per month: ≈ R25,000–R43,000

3. Revenue Estimate

Let's assume:

- **Throughput:** 300 objects/min (18,000/hour)
- **Operational hours:** 4 hours/day × 20 days/month = 80 hrs/month
- **Average value per sorted item:** R0.50 (mixed recyclables blend, conservative)

$$\text{Monthly revenue} = 18{,}000 \times 80 \times R0.50 = R720{,}000$$

4. Profit Calculation

- **Gross margin** = R720,000 – OPEX (~R35,000 average)
- **Monthly net** ≈ **R685,000**
- **Payback time** (on R1,000,000 CAPEX) ≈ **1.5 months** in this aggressive scenario.

5 Aug

Investigated security frameworks and IoT integration for industrial waste sorting systems.
Research Papers Reviewed:
1. ISA/IEC 62443 series standards documentation
2. Industrial cybersecurity threat landscape reports (2024)
3. Azure Machine Learning federated learning capabilities

Cybersecurity Analysis:
98% of IoT traffic remains unencrypted in industrial settings
75% of infected devices serve as network entry points
Zero-trust architecture with micro-segmentation essential

Security Implementation:
Fortinet FortiGate firewalls: Support for 70+ industrial protocols
TLS 1.3 encryption with AES-256 standards
Role-based authentication with multi-factor verification
Network segmentation: OT/IT isolation with anomaly detection

Cloud Integration Strategy:
Microsoft Azure Machine Learning for federated learning
Local PostgreSQL databases for POPIA compliance
Data masking for facility operational details protection
Automated lifecycle policies for low-confidence data removal

Time : 6 hours
Tasks : Security architecture design, compliance framework

6 Aug

Conducted detailed financial analysis and market research for South African deployment.

Market Research Findings:
Current recycling facility costs: R4-12M for TOMRA systems
Material values: Aluminum R9.91/kg, Steel R6.50-7.00/kg, PET 50-100% of virgin resin prices
Labor costs in South Africa: R180,000-250,000 per manual sorter annually
Grid instability: Load shedding affects 65% of industrial operations

Financial Model Development:

Capital investment: R3.4M for complete system
Annual operational costs: R4.0M (including 10 personnel)
Revenue potential: R4.7M-9.6M annually (50%-100% recyclable yield assumptions)
Payback period: 8 months (ideal) to 4 years 11 months (conservative)

System	Cost (ZAR)	Accuracy	Throughput	Maintenance
TOMRA AUTOSORT	R8-12M	98%+	High	Moderate
Bühler SORTEX	R4-8M	Good	Very High	Low
AMP Robotics	R6-10M	High	Moderate	High
Proposed System	R3.4M	98%	180/min	Moderate

Time 6 Hrs

Complted Financial modeling, competitive analysis, ROI calculations

Investigated regulatory landscape and certification requirements for industrial deployment in South Africa.

South African Regulatory Framework:
National Environmental Management: Waste Act (2008):
Waste facility licensing requirements
Environmental impact assessment protocols
Operational compliance monitoring
Annual reporting obligations

Certification Requirements:
CE marking for imported equipment (European compliance)
SABS certification for local market acceptance
Electrical safety certification (SANS 10142-1)
Environmental management system certification (ISO 14001)

Compliance Timeline:
Months 1-2: Environmental impact assessment submission
Months 2-4: Facility licensing application and approval
Months 3-5: Equipment certification and testing
Months 4-6: Operational permits and final approvals

Estimated Compliance Costs:
Environmental assessments: R150,000-R300,000
Licensing and permits: R50,000-R100,000
Equipment certification: R200,000-R400,000
Legal and consulting fees: R300,000-R500,000
Total: R700,000-R1,300,000

Occupational Health and Safety Act (1993):
Workplace safety requirements for automated systems
Risk assessment and management protocols
Employee training and certification requirements
Emergency response procedures

POPIA (Protection of Personal Information Act):
Data collection and processing requirements
Cross-border data transfer restrictions
Individual privacy rights and consent
Breach notification protocols

SANS Standards:
SANS 10366: Recycling facility design and operation

7 Aug

Analysed environmental benefits and lifecycle assessment of automated sorting systems.

Research Papers Reviewed:
1. European Environment Agency (2023) - "Waste and recycling environmental impact analysis"
2. Khandelwal, H. et al. (2019) - "Life cycle assessment of municipal solid waste management options"
3. Johnson, J. et al. (2008) - "Energy benefit of stainless steel recycling"

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Environmental Impact Analysis:

GHG emission reduction: 53% compared to landfilling
Energy savings: 75% for recycled metals, 60% for recycled plastics vs. virgin production
Landfill diversion: Up to 80% of processed materials
Water quality protection through reduced leachate generation

POPIA (Protection of Personal Information Act):
Data collection and processing requirements
Cross-border data transfer restrictions
Individual privacy rights and consent
Breach notification protocols

SANS Standards:
SANS 10366: Recycling facility design and operation
SANS 10142: Conveyor systems safety requirements
SANS 1518: Electrical installations for industrial facilities
SANS 10089: Environmental management systems

Sustainability Metrics:

Carbon footprint reduction: ~2,800 tonnes CO2/year for medium facility
Resource conservation: 5,220 tonnes material recovery annually
Energy offset from solar PV: 25kW array + 100kWh storage
Worker safety improvement: Elimination of manual exposure to hazardous materials

Social Impact Assessment:

Job transition: Manual to technical supervision roles
Skills development: Training programs for automation technologies
Community benefits: Reduced environmental health impacts
Economic development: Value streams supporting local manufacturing

Architecture documentation, specification development

Time 7 ours

8 Aug

Detailed investigation of sensor fusion methodologies and calibration procedures.

Research Papers Reviewed:

1. Ghaffari, M. et al. (2023) - "Systematic reduction of hyperspectral images for high-throughput plastic characterization"
2. Serranti, S. et al. (2012) - "Classification of polyolefins using NIR hyperspectral imaging"

Sensor Fusion Algorithm Development: Developed mathematical framework for confidence weighting:
Where $\alpha + \beta + \gamma = 1$, with dynamic weight adjustment based on:

Signal quality metrics (SNR, illumination uniformity)
Historical accuracy performance
Spatial correlation between sensor outputs

Calibration Procedures:

Daily reference material validation
Weekly spectral library updates
Monthly full system recalibration
Automated drift detection and correction

Time : 8 hours
Tasks Completed: Sensor fusion algorithm design, calibration protocol development

9 - 10 Aug - Break

NEW Battery !

11 August

Material Handeling

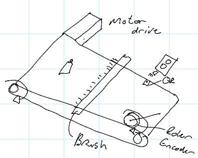
Technical Specs :

Conveyor Speed 2m/S >100 obj/min
Belt - Durable
- Gripy

Polyurit / Rubber / Vynl

En Coder for Position

input : Shaker / Vibrator



Algorythms : Kalman Filter & RRT Path Planning

6 Hrs Conveyor System design

Sensors and Optics:

Primary: Basler (RGB cameras) - German company, SA distributor available
Primary: Specim (NIR hyperspectral) - Finnish company, requires import
Backup: Headwall Photonics (NIR alternative) - US company
Local Support: Specialized distributors in Johannesburg and Cape Town

Processing Hardware:

Primary: NVIDIA (GPU systems) - Global availability, SA support
Primary: Intel/Dell (Server systems) - Strong SA presence
Local Integration: Bytes Technology, Mustek, or similar systems integrators

Robotics and Automation:

Primary: Universal Robots (UR series) - Danish company, SA distributor
Alternative: ABB Robotics - Strong SA manufacturing presence
Local Support: Motion Control Products, Bearing Man Group
Service Contracts: Comprehensive support agreements

Industrial Control:

Primary: Beckhoff (EtherCAT) - German company, SA office in Johannesburg
Alternative: Siemens (Profinet) - Strong SA industrial presence
Local Integration: Control Techniques, Actom, or similar
Supply Chain Risk Assessment:
Currency Risk: 60% of components in foreign currency
Lead Times: 8-16 weeks for specialized sensors
Local Content: 25% achievable with local manufacturing partnerships
Service Support: Critical for 24/7 operations

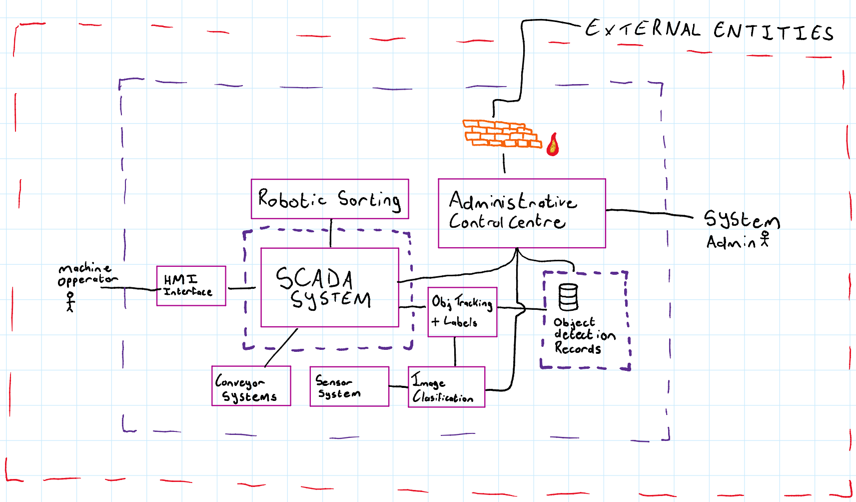
Vendor Management Strategy:

Tier 1 Suppliers: Direct relationships with critical component manufacturers
Tier 2 Suppliers: Local distributors and system integrators
Service Partners: Local companies for installation and maintenance
Backup Sources: Alternative suppliers for critical components

Supply Chain Optimization:
Inventory Management: 3-month spare parts inventory minimum
Local Partnerships: Joint ventures for component assembly
Training Programs: Local technician certification initiatives

12 August - Design Survey

SYSTEM DFD



① SENSOR SELECTION

1.1 Visible Light

1.2 NIR Spectroscopy

X-RAY Imaging

MAGNEIC field

① SENSOR SELECTION

1.1 Visible Light
- ✓ Low Cost
- ✓ Acceptable Objdetection (≥ 80%)
  - ↳ Colour Detection
  - ↳ Shape Detection
- ✓ Achieveable High throughput (60-90FPS ~ 100obj/sec)
- ✓ Scalable
- ✗ Cant distinguish Reflective/Transparent Obj
- ✗ Limited Composition Analysis
- ✗ Lighting dependant (Perifenals)
- ✓ Comercialy Available
- ✓ Many training dataset

✓

1.2 NIR Spectroscopy
- ✓ Good Material Composition Analysis
  - ↳ Polymer Clasification (Future improve)
- ✓ Transparency Handeling
- ✗ High Cost
- ✗ Slower DATA Aquisition
  - ↳ (10-40 FPS ~ 10 Obj/sec or 2000bj/min)
- ✗ Lighting dependant
- ✓ Comercially Available
- ✓ DataSet Available for training

✓

X-RAY Imaging
- ✓ Good Material Composition
- ✗ Safety Hazard
- ✗ High Power
- ✗ High Cost
- ✓ Available (TSA Scaner)
- ✓ DATA Set Avail
- Slow DATA Aquisition

✗

MAGNEIC field
- ✓ Cheap
- ✓ Remove metals

line-scan NIR or point or full-area NIR

Feature	Point NIR	Line-Scan NIR ✓	Snapshot Hyperspectral NIR
Field of view	1 spot	Entire belt width	Entire belt width
Speed	~8.7 Hz	300–1000 lines/sec	5–30 fps
Best for	Isolated items	Continuous conveyor sorting	Static or slow-moving scenes
Complexity	Low	Medium	High
Throughput potential	<150 items/min	300–500+ items/min	100–200 items/min

9Hrs

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13 Aug   Survey Continued

Sensor Intergration

NIR + RGB

Confidence scoreing from each Clasification Algo.
Conveyor ENCOder to match RGB & NIR Pixel values together

Implementation Plan

Mounting & Alignment
- the line-scan NIR camera above the conveyor so each captured line covers the full belt width.
- Mount uniform NIR lighting on both sides at ~45° angle.

Conveyor Synchronization
- Use an **encoder** to track belt movement and trigger each line capture so the vertical resolution matches your desired mm/line.
- E.g., 2 mm spatial resolution → at 2 m/s belt speed, need 1000 lines/sec.

RGB + NIR Fusion
- Keep existing RGB camera for shape/color detection (90 FPS).
- Use object tracking from RGB to segment the NIR line-scan data and match spectral signatures to each object.

Data Processing
- Preprocess spectral lines into a **2D+λ hyperspectral cube** per object.
- Reduce bands (via PCA or selecting key wavelengths) to lower computational load.
- Feed into a **1D CNN** or SVM classifier for material detection.

Decision & Sorting
- Fuse RGB and NIR classification results using your existing weighted confidence system (5 ms fusion time is fine).

NIR - 1D λ intensity

Trigger Coordination   Line scanning

if YOLO Certanty score Low ∴ Trigger NIR Scan

- Trigger pneumatic sorters with predicted drop timing.

< 0% Combined detection Rate

② AI Object Detection & Material Classification

2.1 Yolo v11
- > 25 ms inferenc
- > WasteNet Data for preliminary training
- > Active learning
- > i/p: - 640×640 Pixels
  - ↳ each with RGB Values

2.2 CNN - 1D
- > 15 MS inference
- >
- > Active learning
- > i/p: Pixel Array
  - 256 wavelength intensity

Sliding Window

5HRS

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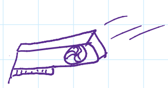
14 August

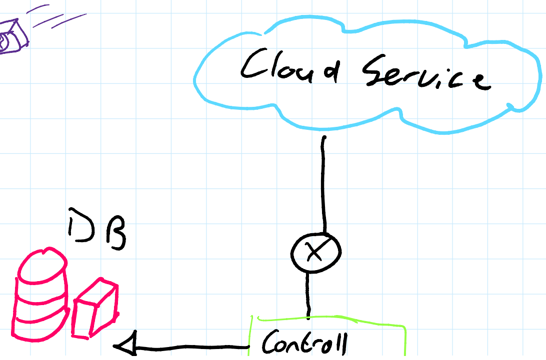
Server Tech Stack

DB
- Space needs
- Access
- Security
- POPI

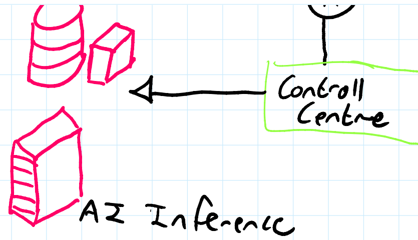
AI
- Process heavy
- Low Latency
- 10Gig EThen
- GPU - A 6000
- 512 RAM DDR5 ECC



- > X86 64/128 XEON
- > POSGres / Mongo
- > 100 TB
- > HDD
- > SSD / Nvme Cache
- > 10 Gig EThn
- > 128 DDR5 ECC



> SSD / Nvme Cache
> 10 Gig Ethn.
> 128 DDR5 ECC



Hewlett Packard Enterprise

BREAKDOWN OF COMMON RAID LEVELS

RAID LEVEL	METHOD	HARDWARE / SOFTWARE	MINIMUM # OF DISKS	COMMON USAGE	PROS	CONS
JBOD	SPANNING		2	INCREASE CAPACITY	COST-EFFECTIVE STORAGE	NO PERFORMANCE OR SECURITY BENEFITS
0	STRIPING		2	HEAVY READ OPERATIONS	HIGH PERFORMANCE (SPEED)	DATA IS LOST IF ONE DISK FAILS
1	MIRRORING		2	STANDARD APP SERVERS	FAULT TOLERANCE, HIGH READ PERFORMANCE	LAG FOR WRITE OPS, REDUCED STORAGE (BY 1/2)
5	STRIPING & PARITY		3	NORMAL FILE STORAGE & APP SERVERS	SPEED + FAULT TOLERANCE	LAG FOR WRITE OPS, REDUCED STORAGE (BY 1/3)
6	STRIPING & DOUBLE PARITY		4	LARGE FILE STORAGE & APP SERVERS	EXTRA LEVEL OF REDUNDANCY, HIGH READ PERFORMANCE	LOW WRITE PERFORMANCE, REDUCED STORAGE (BY 2/5)
10 (1+0)	STRIPING & MIRRORING		4	HIGHLY UTILIZED DATABASE SERVERS	WRITE PERFORMANCE + STRONG FAULT TOLERANCE	REDUCED STORAGE (1/2), LIMITED SCALABILITY

What Happened to 2-4 and 6-9?

The RAID levels described above are the most common levels used in enterprise scenarios. The levels in between are highly specialized and only make sense in very specific scenarios.

7 Hrs

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15 August

Research colation

Reviewed Collection of Research Papers
↳ Selected ones Relevant to my design

Implementation Phases:

1. Phase 1 (Months 1-3): Component procurement and facility preparation

2. Phase 2 (Months 4-6): System installation and initial testing
3. Phase 3 (Months 7-9): Integration testing and optimization
4. Phase 4 (Months 10-12): Production deployment and training

Stakeholder Engagement:
Municipal authorities: Permits and regulatory compliance
Waste management companies: Pilot implementation partnerships
Research institutions: Ongoing development collaboration
Community organizations: Social impact assessment and engagement

Resource Requirements:

Project team: 8-12 professionals (engineers, technicians, project managers)
Facility requirements: 200m² minimum floor space
Utilities: 50kW electrical capacity, compressed air, network infrastructure
Training programs: 40 hours initial training, 20 hours annual updates

Success Metrics:
Technical performance: 98% sorting accuracy, 180 items/min throughput
Economic performance: ROI within 5 years, operational cost reduction
Environmental impact: 80% landfill diversion, 2,800 tonnes CO2 reduction/year
Social impact: Job creation in technical roles, worker safety improvement

8 Hrs

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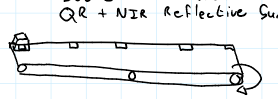
19 August — Revisit mechanical System

3 Conveyor System

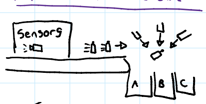
Belt driven
1024 PPR Encoder (for Position)
Steper servo motor
QR + NIR Reflective Surface

Kallman Filter ✗



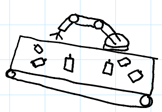
4 Sorting System

4.1 Pnumatic Blower



✓ Fast
✗ High unpredictability
↳ Varying MASS
↳ Varing bouncyness
✗ Fragile items!
✓ Cheap

4.2 Robotic Arm



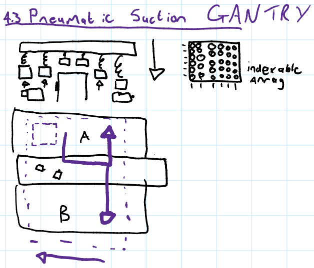
✗ Slow (1 item Per Pick)
✓ Fragile items
✓ Handes Varying Size
✗ Costly

Sensors and Optics:
Primary: Basler (RGB cameras) - German company, SA distributor available
Primary: Specim (NIR hyperspectral) - Finnish company, requires import
Backup: Headwall Photonics (NIR alternative) - US company
Local Support: Specialized distributors in Johannesburg and Cape Town

Processing Hardware:
Primary: NVIDIA (GPU systems) - Global availability, SA support
Primary: Intel/Dell (Server systems) - Strong SA presence
Local Integration: Bytes Technology, Mustek, or similar systems integrators

Robotics and Automation:
Primary: Universal Robots (UR series) - Danish company, SA distributor
Alternative: ABB Robotics - Strong SA manufacturing presence
Local Support: Motion Control Products, Bearing Man Group

4.3 Pneumatic Suction GANTRY



✗ Complex Control System
✓ fast Item Processing
✓ Medium Cost

Industrial Control:
Primary: Beckhoff (EtherCAT) - German company, SA office in Johannesburg
Alternative: Siemens (Profinet) - Strong SA industrial presence
Local Integration: Control Techniques, Actom, or similar

Supply Chain Risk Assessment:
Currency Risk: 60% of components in foreign currency
Lead Times: 8-16 weeks for specialized sensors
Local Content: 25% achievable with local manufacturing partnerships
Service Support: Critical for 24/7 operations

Vendor Management Strategy:
Tier 1 Suppliers: Direct relationships with critical component manufacturers
Tier 2 Suppliers: Local distributors and system integrators
Service Partners: Local companies for installation and maintenance
Backup Sources: Alternative suppliers for critical components

Supply Chain :
Inventory Management: 3-month spare parts inventory minimum
Local Partnerships: Joint ventures for component assembly
Training Programs: Local technician certification initiatives
Service Contracts: Comprehensive support agreement

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20 August

CYBER SECURITY

> NEWORK Segmentation (DMZ) VLAN Subnetting
> FIRE WALL (Fortinet FortiGate)

> Role based authentication (Certificate device Auth)

> DATA Protection (AES 256 + TLS 1.3 - transit data)

> Network monitoring for Anomalies transmit Packets

SYSTEM Connectivity

> Local AI Processing
> VPN Tunneling
> SAP Intergration (Process Monitoring)

SCADA
> PROFINET

DB
> SQL Server

5 HRS

21 August

Conducted comprehensive risk analysis for industrial deployment.

Technical Risks:
Sensor degradation under industrial conditions
Software reliability for 24/7 operation
Integration complexity between subsystems
Cybersecurity vulnerabilities in connected systems

Mitigation Strategies:
Redundant sensor systems for critical functions
Comprehensive testing protocols before deployment
Staged rollout approach for risk management
Regular security audits and updates

Operational Risks:
Material supply variability affecting system performance
Operator training and skill development requirements
Maintenance complexity and parts availability
Economic factors affecting project viability

Financial Risks:
Currency fluctuation impact on imported components
Competition from established international systems
Changes in recycling material market values
Regulatory changes affecting compliance costs

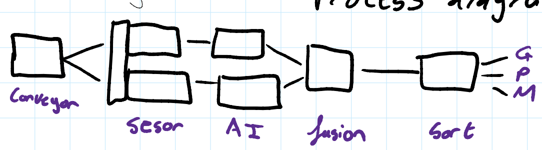
Risk Matrix Development:
High/Medium/Low probability assessment
Impact severity evaluation
Mitigation cost analysis
Contingency planning for critical risks

9 Hours

25 August

Report writing

Process diagram



> Developed Introduction Context
> Developed Report back Ground
↳ SA Context - Price + waste
↳ AI Wast Sorting - Yolo CNN
↳ Mechanical Sorting - Pneumatic - Robotic
↳ IoT + Cyber Security - Firewall - TLS

> finished Section 2 + Edited Section 1
> Drafted Section 3 High level Architecture
> Edit Section 2
> Created high level diagram

9 s hr

26 August

Report writing:

Section 4:

> Detailed Design
↳ Operational Support
↳ DFD
↳ Process flow
↳ Server Archi - Local + Cloud
> Appendix (Tech Stack

> Mechanical Layer
↳ Conveyor - 7kg 8KW
↳ Sorting - Robotic Arm - Moveit
↳ Encoder - optical

↳ Controll Layer
↳ Sync - Master Clock
↳ SCADA -
↳ User input - HMI SCADA Applic.
↳ Protocol - EtherCat

↳ AI inference
↳ Take Comp. - CV & Spectral Analysis

finish Tomm.

11 Hr

11 Hr

27 August

Report writing
> Edit Section 2 from yesterday
> finish AI Inference
  ↳ Sensor fusion
> Added Image explination
> Data Acquisition Layer
  ↳ Compare Sensors
    ↳ Already in section 2
  ↳ RGB Sensor
    ↳ Specs
    ↳ Comparison
  ↳ NIR Sensor
    ↳ Specs
    ↳ Comparison
> Edit section 2
> Started System evaluation
  ↳ throughput
  ↳ Power usage

8 Hr

28 Aug

Report writing
> Edit section 4
> format tabels
> Finished System Eval.
> Calculated + Discussed financial analysis + Appendix B
  ↳ OPEX + CAPEX
> Discussed Sustainability

> Impact Analysis
> Enviro. Complete
  ↳ Land fill
  ↳ water leeching
  ↳ CO_2
> Social Complete
  ↳ Job Loss
  ↳ Social Transform
  ↳ upskill
> Economic Tomorrow!
  ↳ Circular Econ.
  ↳ Capex
  ↳ investments

> Future Improvements added

9 Hr

29 Aug

Report writing
> Edit whole Report : fixed flow
> Completed Impact section
> Add Refferences
> formated Tabels
> Draft Abstract and Conclusion
> formated headings

10 pt

2 Left; Right, bottom
3 Top

8 hrs

29 August

Report writing
> Finished Abstract and Conclusion
> Proof Read

Report Writing

> Finished Abstract and Conclusion

> Proof Read

> Started Non Technical Report

7 Hrs

30 Aug

> Finished Non Tecchnical Report 11pt 2pg

> Added Refferences

> Proof Read

6 HRS

31 Aug

> Proof Read Technical Report

> Proof Read Non-Techical Report

> Created Cover Page

4 Hours

Submission Tommorow!

@ 8:00